Hidden Patterns in Customer Purchasing Behavior

Executive Summary

This analysis examines hidden patterns in customer purchasing behavior using customer lifetime value (CLTV), order activity, discounts, app event behavior, and campaign performance. Across **22,955 customers**, the business produced about **$5.7M in total lifetime value** from **628,531 orders**, but the customer base does not behave like a typical marketplace where many customers contribute small amounts. Instead, purchasing behavior is extremely uneven. The data shows that **93.4% of customers have never purchased**, and among those who do purchase, the business depends heavily on a very small elite segment. Discount behavior also shows a paradox: the customers who use the most discounts fall into two extremes high value power users who are not actually dependent on discounts, and deal hunters who generate almost no value. Funnel analysis suggests meaningful drop offs at entry and checkout, while campaign analysis indicates promotions are underperforming and may not be reaching the right audience. Overall, the hidden pattern is that the business is running on a narrow high value core, surrounded by a large inactive majority and a large one time buyer population that fails to convert into repeat customers.

Inactivity Crisis and Database Structure

A major hidden pattern is inactivity. Out of **22,955 customers**, **21,450 (93.4%)** have never made a purchase, leaving only **1,505 active customers**. This creates a distorted view of the customer base because the database looks large, but the true purchasing population is much smaller. This inactivity can mean two very different realities. It could indicate poor conversion and weak activation strategy, where most users sign up but never transition to a first order. However, it could also reflect data quality issues such as duplicate records, test accounts, bots, or outdated historical entries that are still counted as customers. This is important because it affects every metric. If the majority of the database is non real or non convertible, acquisition and retention KPIs will look artificially weak, and targeting campaigns become less effective because the "audience pool" includes many users who are not actually reachable buyers.

Extreme Revenue Concentration Beyond the 80/20 Rule

Among active customers, revenue concentration is unusually extreme. The top **10% of active customers generate 90.5% of total revenue**, and the top **20% generate 95.1%**, meaning the business follows a 90/10 pattern rather than a typical Pareto distribution. Numerically, the top **150 customers contribute around $4.5M in CLTV**, while the bottom **80% of active customers contribute only about $245K**. This structure signals high dependency risk because losing a small number of top customers could significantly reduce revenue. Order frequency patterns support the idea that value is driven mainly by repetition rather than high single purchases. The **median number of orders is 1**, meaning half of active customers buy only once, but the **mean is 417.6**, which is inflated by power users. A 95th percentile around **1,500 orders** shows that a small group places extremely high numbers of orders. This skew is a strong hidden pattern because it means the business is not “balanced”; it is closer to being a power user driven ecosystem.

Discount Paradox and Segment Behavior

Discount analysis shows what looks like a contradiction at first. High discount users have an average **$9,757 CLTV**, while low discount users average **$2,314 CLTV**, a significant gap of **$7,443 ($p < 0.001$)**. But this does not mean discounts “create” loyalty. The explanation is hidden inside segmentation. Two groups behave very differently but both appear in the “discount users” category. One group, **High Value Deal Seekers (277 customers)**, has an average **$16,922 CLTV**, about **1,176 orders**, and only **3% discount dependency**. These users seem to capture discounts when available, but their spending is not driven by promotions. The second group, **Deal Hunters (308 customers)**, has an average CLTV of only **$19.79**, an average order value of **$8**, and a discount dependency of **332%**, meaning discounts dominate their purchasing behavior and they generate close to no value. This reveals a hidden business inefficiency: discounts can simultaneously reward customers who would buy anyway and attract customers who are not profitable. Although the total discounts given are **$375,428** against **$4,970,739 revenue** (suggesting a 13.24× ratio), the real concern is that much of this discount spending may be margin loss rather than incremental growth. If even 50% of discounts go to customers who would purchase without them, the implied hidden cost becomes roughly **$187K–$200K**, which is significant for profitability.

The One Timer Trap and the Second Purchase Threshold

A major behavioral pattern is that **841 of the 1,505 active customers (55.9%)** make exactly one purchase and disappear. This group has an average CLTV of **$292** and contributes around **$246K**, or only **4.9% of revenue**. In contrast, repeat customers (664 buyers) average **$7,482 CLTV** and contribute **$4.7M**, which is **95% of revenue**. The gap suggests that loyalty is not formed at the first purchase. The real threshold is the second purchase. If the company cannot move a buyer to purchase twice within a short window, they likely remain a low value customer permanently.

Digital Funnel Patterns and Purchase Intent

The app event path analysis shows two major drop off points. Around **40% of users leave immediately after app_start without viewing a place**, and around **34% of users abandon their cart before submitting an order**. At the same time, there is evidence of "quick shoppers," where users add items directly to cart without browsing, suggesting a loyal group that uses quick add or repeat ordering behaviors. The basket building pattern is also clear: users add about **4 items** per cart and view a place about **1.6 times** before ordering, which suggests decision making behavior like price checking or confirming options. This shows that digital behavior is not uniform. New or casual users struggle to engage, but experienced users move fast. This implies the app experience might work well for repeat customers but fails to convert or retain new ones.

Campaign Performance and Cannibalization Risk

Campaign performance is relatively weak, with an overall redemption rate of **2.51%** across **641 campaigns**, where only **2,956 redemptions** were used out of **117,713 available**. The most effective approach appears to be item specific campaigns, which achieve about **8.75% redemption**, and discounts between **21–30%**, which reach **9.46% redemption**. Deep discounts above **50% show 0% redemption**, meaning high discount depth is not generating uptake. Urgency matters strongly: campaigns lasting less than one day perform much better than multi week campaigns, showing a clear "flash sale" effect. Cannibalization appears low right now because campaigns are underused, but risk increases if campaigns become effective without improving targeting. For example, a premium item priced at **$280** had **14 redemptions out of 34 purchases** under a **30% discount**, implying that about **41%** of its purchases were discounted and likely would have happened anyway, costing about **$1,176 in lost revenue**.

Limitations and Areas That Need Further Investigation

Even though the analysis reveals strong patterns, there are important limitations that affect how confidently we can interpret some results. One limitation is that some datasets appear incomplete, sparse, or not aligned in a way that supports full customer journey tracking. For example, campaign redemption data is available at the aggregate level, but without full customer exposure logs (who actually saw a campaign, clicked it, and then purchased), the study cannot measure true incremental lift. A redemption rate of 2.51% might indicate failure, but it could also be influenced by limited campaign visibility or missing exposure tracking. Another limitation is the presence of values that look unrealistic or suspiciously extreme. The mean order count of **417.6** and the observation that some users place around **1,500 orders** can be real in a subscription like system, but it could also be inflated by commercial accounts, internal users, franchise operators, or duplicated activity logs. This becomes even more likely because **only 6 users created all 641 campaigns**, which is unusual for a normal customer base and suggests employee accounts, franchise owners, or B2B operators. If these users are mixed into the “customer” population, CLTV and frequency statistics become biased upward and make the business appear more power user driven than it truly is.

A third limitation is that large inactivity (93.4%) could be partially artificial. It may include test accounts, duplicates, invalid sign ups, legacy profiles, or users from older systems. Without a proper account audit and identity matching, we cannot conclude that all inactive users represent real potential customers.

Finally, the campaign cannibalization assessment is limited because purchase time series at item level is incomplete for before/after comparisons. Without clean purchase timestamps linked to campaign periods at the customer and item level, it is not possible to precisely estimate how many discounted purchases are incremental versus cannibalized. The study can identify risk signals, but not final causal impact. Because of these limitations, the results should be treated as strong directional findings, but some metrics especially the extreme order counts, campaign creator patterns, and inactivity magnitude should be validated through a deeper data audit.

Conclusion

The hidden patterns in customer purchasing behavior show that the business depends on a tiny group of high value repeat buyers, while most users are inactive or one time buyers. Discounts do not consistently build loyalty and often benefit customers who would buy anyway or attract deal hunters who generate little value. Digital behavior highlights engagement and checkout friction, and campaign data suggests promotions are currently underperforming but show signs that targeted moderate discounts and urgency based strategies can work better. However, data limitations and potential anomalies, including empty or incomplete datasets, unrealistic outlier behavior, and suspicious campaign creation patterns, mean that the next step should involve a data quality audit and improved tracking. After validating the customer base and ensuring clean journey level data, the business can confidently implement retention focused strategies, reduce unnecessary discounting, and design better campaign targeting that drives repeat behavior rather than one time purchases.